

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO3: Articulation (BL4, BL5)	Assessment:	Assessment Tool 2 – Case Study
Weightage:	20% of CIA	Total Marks:	100
CO3 Statement:	Design Divide and Conquer algorithms for Merge Sort, Quick Sort, Binary Search and Matrix Multiplication		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Carefully analyze the given case study questions.
- Explain in detail with sample code if needed.
- Clearly identify all key issues, provide an in-depth analysis, and propose innovative, feasible solutions with strong justification.

Question Type	Focus Area	Questions
Case Study	Focus on Design Divide and Conquer algorithms: Merge Sort, Quick Sort, Binary Search and Matrix Multiplication	Q1

SECTION A – CASE STUDY

**SIMATS**
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SIMATS Online Programming Lab
DAA PROGRAMMING

JENIECATHRINA J
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CO3 AT2-CASE STUDY

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Language: Python C C++ Java

QUESTIONS

Q1 Weather Simulation Model (Matrix Multiplication) A weather simulation system performs repeated matrix operations for prediction 100 marks

Analyze how divide and conquer matrix multiplication improves performance. Identify computational challenges. Apply optimized techniques and analyze complexity. Design a solution and justify its efficiency.

Your Code

```
1 def add_matrix(A, B):
2     n = len(A)
3     return [[A[i][j] + B[i][j] for j in range(n)] for i in range(n)]
4
5
6 def subtract_matrix(A, B):
7     n = len(A)
8     return [[A[i][j] - B[i][j] for j in range(n)] for i in range(n)]
9
10
11 def split_matrix(A):
12     n = len(A)
13     mid = n // 2
14
15     A11 = [row[:mid] for row in A[:mid]]
16     A12 = [row[mid:] for row in A[:mid]]
17     A21 = [row[:mid] for row in A[mid:]]
18     A22 = [row[mid:] for row in A[mid:]]
19
20     return A11, A12, A21, A22
```

Input

stdin input

Output

Code of Conduct

I jeniecathrina 192311229(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: jeniecathrina

RUBRICS FOR CALCULATION

Case Study					
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25%	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20%	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20%	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10%	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25%				
Application of Concepts	25%				
Analysis	20%				
Solution Design	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____